# Lecture 7: Unions

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# What are trade unions?

Trade unions are organisations of workers whose primary objectives are to improve the pecuniary and non-pecuniary conditions of employment among their members.

Unions bargain with employers on virtually all aspects of the employment contract:

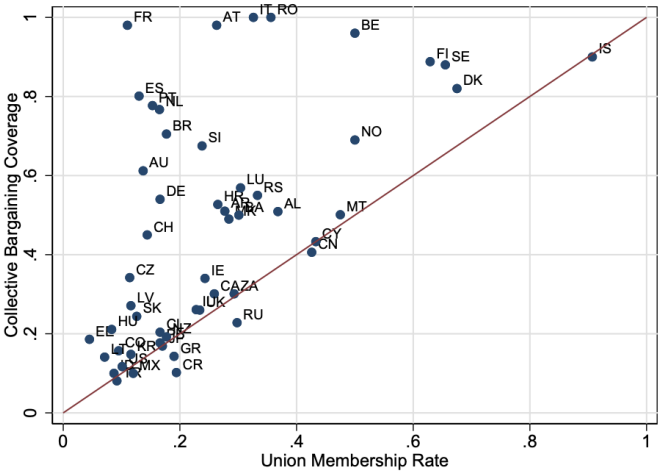
- Pay and employee benefits
- Conditions of work
- Hiring, job assignment, promotion, layoff

Unionisation largely differs across countries along may dimensions.

# Key institutional dimensions

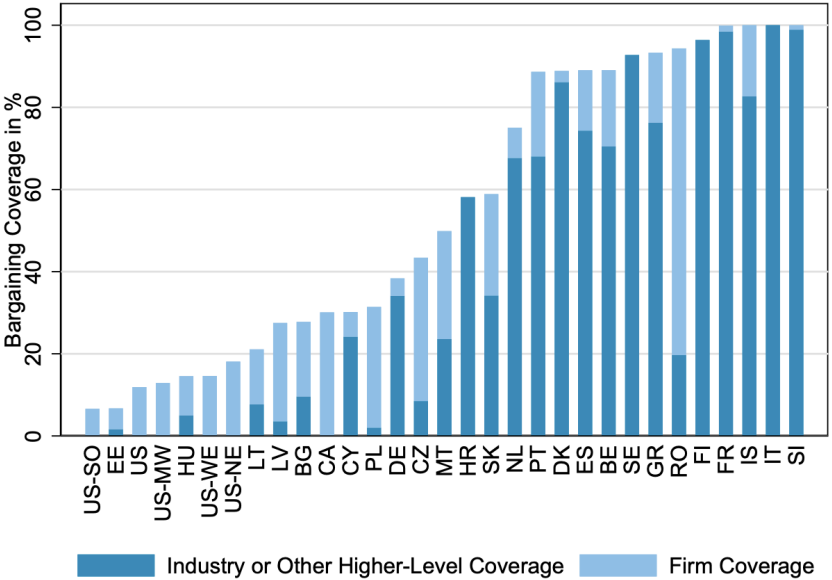
- **Union density:** the fraction of workers registered with a trade union.
- **Bargaining coverage:** the percentage of the eligible workforce whose contract is regulated by collective agreements.
- **Bargaining (de)centralisation:** the level (plant, firm, industry, country) at which contracts are negotiated.
- **Coordination:** formal centralisation versus implicit (tacit) coordination between independent unions and employers.

# Coverage often far exceeds density




Source: Jager, Naidu and Schoefer (2024)

# Coverage varies greatly across countries



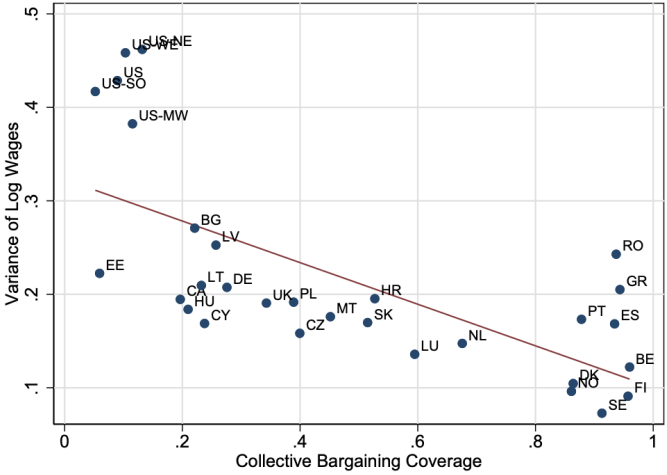
# Institutional arrangements

Three forms of shop arrangement determine the union's ability to act as a monopoly supplier of labour:

- **Open shop:** Workers may be employed without joining the union.
- **Closed shop:** Workers must be union members to be employed (assumed in most formal models).
- **Union shop:** Non-union workers may be hired but must join the union within a brief period.

A union's bargaining power rests primarily on how credible the **threat of strike** is and on preventing replacement of striking workers (hence density matters).

# Why do we care? Union coverage and wage inequality

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# Roadmap

- What do unions do?
- Why an union pay premium?
- Beyond pay premium

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- **What do unions do?**
- Why an union pay premium?
- Beyond pay premium

# What do unions do?

1. Monopoly union view
2. Efficient bargaining view
3. Rent extraction (holdup) view
4. “Voice” view

These views are not mutually exclusive. They may coexist and operate under different conditions.

# What do unions do?

1. **Monopoly union view**
2. Efficient bargaining view
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# Monopoly union model: Setup

According to the standard view, unions are frequently able to push wages above the competitive level (hence the “monopoly role” of unions).

The simplest model of union wage-setting:

- The **union moves first** and sets the wage $w$.
- The **firm moves second** and chooses employment $L$, taking the wage as given.

**Timing rationale:** Unions can influence wages, while firms retain discretion over hiring. This resembles the reality of most CBA processes.

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# Firm behaviour given the wage

Firm profits:

$$\pi = R(L) - wL$$

From FOC:

$$R'(L) = w$$

- This implicitly defines the firm's labour demand curve $L^D(w)$.

The slope of the firm's isoprofit curve (holding $\pi$ constant) is:

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# Sketch of the firm's labour demand curve

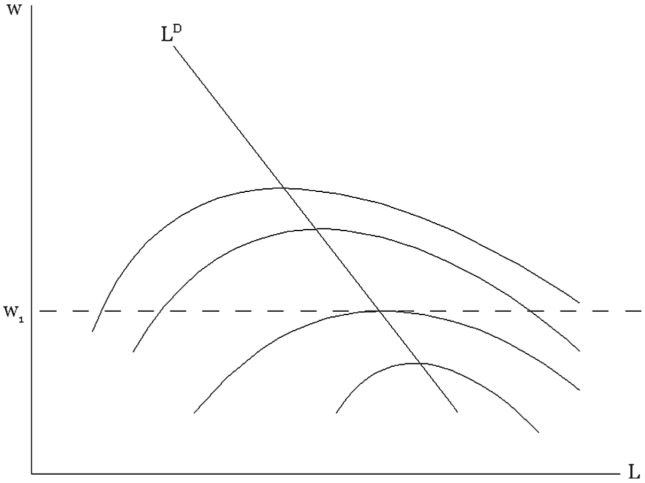



**Fig. 1** Labour demand curve

Source: Lawson (2011)

# Union preferences

As Dunlop (1944) noted, specifying the union's objective is not straightforward.

Unions typically care about:

- higher wages, and
- higher employment for their members.

These trade-offs are embedded in a utility function $U(w, L)$, with $U_w > 0$ and $U_L > 0$.

# Union preferences: Examples

Union utility function $U(w, L)$ can assume different structural forms, which correspond to different union maximisation goals:

- Total wage bill: $U = wL$
- Rent: $U = (w - w^c)L$
- Weighted preferences over wage (rent) and employment: $U = (w - w^c)^\theta L^{1-\theta}$
- Utilitarian: $U = Lu(w) + (t - L)u(b)$
- Expected utility: $U = \frac{L}{t}\, u(w) + \left(1 - \frac{L}{t}\right) u(b)$

in which $w^c$ (or $b$) is the reservation/alternative wage; $t$ is union membership; $w > b$; and $0 < L \leq t$.

# Union indifference map in (L, w) space

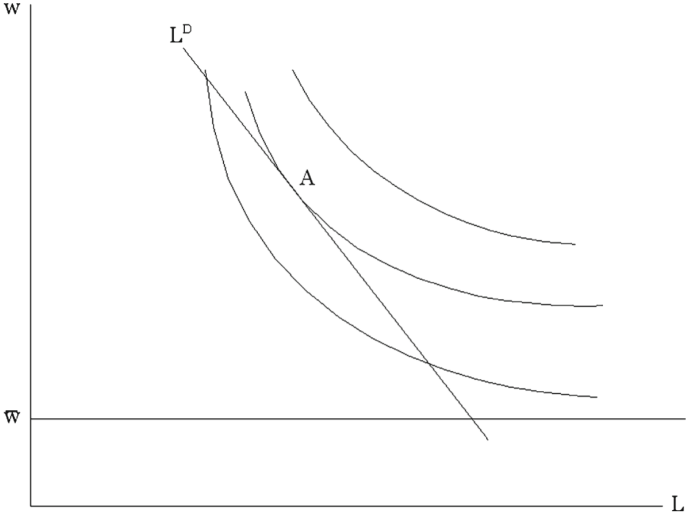



**Fig. 2** Tangency of labour demand curve and union indifference curve

# Monopoly union: Optimisation

The union chooses the wage $w$ to maximize its utility $U(w, L)$, with $U_w > 0$ and $U_L > 0$.

Because the firm responds optimally, this is equivalent to choosing $(w, L)$ subject to:

$$R'(L) = w$$

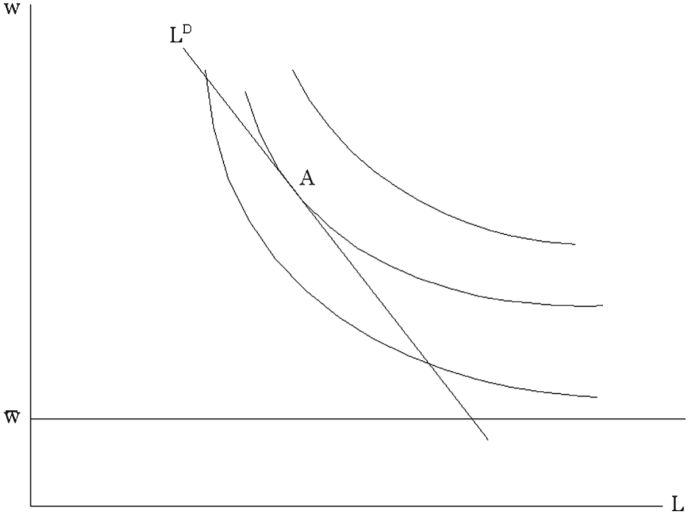
The first-order condition is:

$$-\frac{U_L}{U_w} = R''(L)$$

The union chooses the point where:

- Indifference curve is tangent to the firm's labour demand curve.

# Tangency of labour demand curve and union indifference curve

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# Monopoly union: An example with risk-neutral workers

Using $u(w) = w$, the union solves:

$$\max_{w} \quad U = \frac{L}{t}(w - b) + b \quad \text{s.t.} \quad R'(L) = w$$

The optimality condition reduces to:

$$\varepsilon = \frac{w}{w - b}, \qquad \text{where } \varepsilon = -\frac{w}{L} L'(w)$$

The union equates the elasticity of labour demand to the proportional wage markup over the outside option $b$.

**Comparative statics:**

- Higher $\varepsilon \Rightarrow$ lower $w^*$.
- Higher $b \Rightarrow$ higher $w^*$, lower $L^*$.
- Higher $t$ has no effect on $w^*$ or $L^*$ under this specification.

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# Why the monopoly union model is appealing

The monopoly union model is popular because its structure resembles real-world bargaining.

- Unions often negotiate wages.
- Firms often retain discretion over employment levels.

This makes the model realistic and easy to interpret.

# Inefficiency of the monopoly union outcome

Despite its appeal, the monopoly union outcome is **not Pareto efficient**. Why?

- The chosen outcome lies on the labour demand curve.
- But there exist wage–employment combinations off the labour demand curve that could be Pareto improving.

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# Inefficiency of monopoly union outcome

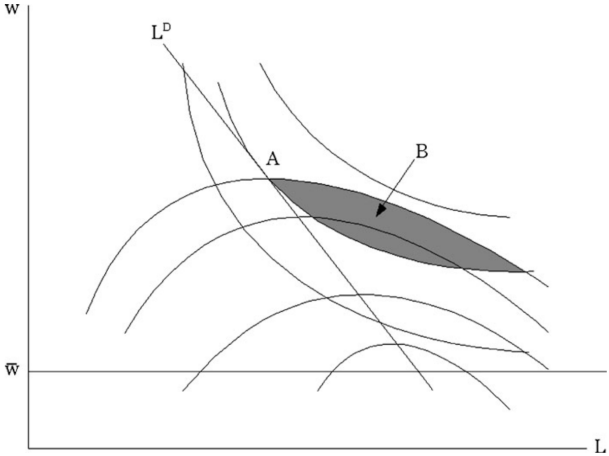


**Fig. 3** Inefficiency of monopoly union outcome

In region B, at least one of the firm and union can be made better off than at point A without making the other worse off.

# What do unions do?

1. Monopoly union view
2. **Efficient bargaining view**
3. Rent extraction (holdup) view
4. “Voice” view

# The efficient bargaining model

To overcome this inefficiency, firm and union can bargain over **both wages and employment** simultaneously.

Also, the monopoly union assumption is too extreme: “The union never gets everything it wants” (Layard et al. 1991).

The Pareto-efficient outcome satisfies:

$$\underbrace{-\frac{U_L}{U_w}}_{\text{slope of IC}} = \underbrace{\frac{R'(L) - w}{L}}_{\text{slope of isoprofit}}$$

The locus of all Pareto-efficient $(w, L)$ pairs is the **contract curve**.

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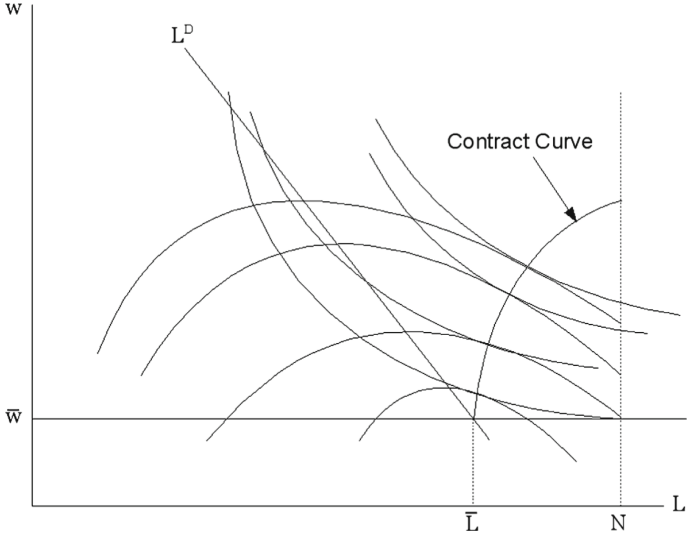


**Fig. 4** Contract curve

Source: Lawson (2011)

# Properties of the contract curve

Relative to the monopoly union outcome:

- **Employment is higher** (and wages are lower).
- Profits **do not necessarily fall** — the firm may be better off.
- Both parties can gain from moving to the contract curve.

**Empirical challenge:** Bargaining over employment is rare in practice. Uncertainty about future product demand can make joint contracts non-incentive-compatible.

# From monopoly union to bargaining

Maybe a more realistic characterisation is:

- Unions and firms **bargain over wages**.
- Firms retain the **right to manage** employment decisions.

This is the **Right-to-Manage (RTM)** model. To solve it, we use **Nash bargaining**.

# Nash bargaining solution

Nash proposed an axiomatic approach. The bargaining outcome solves:

$$\max_{w} \quad \Phi = \underbrace{\left[\frac{L}{t}(w-b)\right]^{\beta}}_{\text{union net gain}} \cdot \underbrace{[\pi(w)]^{1-\beta}}_{\text{firm net gain}}$$

where $\beta \in [0, 1]$ is the union's relative bargaining power.

- $\beta = 1$: Monopoly union outcome
- $\beta = 0$: Perfectly competitive outcome ($w = b$)
- $0 < \beta < 1$: RTM outcome — intermediate

# RTM model: Results

From FOC:

$$\beta\varepsilon + (1-\beta)\varepsilon_\pi = \frac{w}{w-b}$$

where $\varepsilon_\pi = -w\pi'(w)/\pi$ is the elasticity of profit with respect to wages.

**Main implications:**

- All comparative statics from the monopoly union model are confirmed.
- $\uparrow \beta \Rightarrow \uparrow$ wage, $\downarrow$ employment.
- The outcome always lies on the labour demand curve (firm retains hiring discretion).

# What do unions do?

1. Monopoly union view
2. Efficient bargaining view
3. **Rent extraction (holdup) view**
4. “Voice” view

# The holdup problem

The **holdup problem** (Grout, 1984) arises when:

- Long-term contracts are difficult or impossible, and
- Firms make irreversible (sunk) investments in capital.

Problem:

1. Firm invests in capital (sunk cost).
2. Union demands higher wages *after* investment is made.
3. The firm cannot easily withdraw $\Rightarrow$ union captures part of the returns.

Anticipating this, firms **under-invest** relative to the social optimum.

This creates inefficiency because the union's ability to extract rents ex post distorts ex ante investment incentives.

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# The voice view of unions

Freeman and Medoff (1984) emphasize that unions have two opposing effects.

1. **Monopoly face:**
   - Unions can raise wages above the competitive level.
   - This may reduce employment and generate allocative inefficiencies.

   However, in a world of imperfect information and uncertainty, unions can improve efficiency (the "good face" of unions).

2. **Voice face:**
   - Unions provide a collective channel for worker complaints ("voice" instead of "exit")
   - Unions contribute to make implicit contracts (self-)enforcing.
   - This can improve workplace conditions, morale, and productivity.

# Which View is Correct?

There is no single "correct" view of unions. The different models describe different mechanisms:

| View | Key mechanism | Most relevant when... |
|---|---|---|
| Monopoly | Unions raise wages above competitive level via market power | Union has strong wage-setting power |
| Efficient bargaining | Joint negotiation over $w$ and $L$ reaches the contract curve | Parties can commit to employment levels |
| Holdup | Ex post rent extraction deters ex ante investment | Capital is sunk, contracts incomplete |
| Voice | Collective channel improves information, morale, productivity | Turnover costly, information problems important |

The **net effect** of unions is theoretically and empirically **ambiguous**.

# Summary

1. **Monopoly union model** has wages above the competitive level but is not PE.
2. **Right-to-Manage model** generalises via Nash bargaining; outcomes depend on relative bargaining power $\beta$.
3. **Efficient bargaining** over both $w$ and $L$ can reach the contract curve, but is rare.
4. **Holdup problem** distorts investment; **voice view** highlights potential efficiency gains.

# Roadmap

- What do unions do?
- **Why an union pay premium?**
- Beyond pay premium

# Empirics: Why an union pay premium?

Empirically, the literature moved to understanding whether and why there is union pay premia.

Increasingly accepted view in labor economics: differences in productivity across firms "spill-over" onto wages (Van Reenen (1996); Card et al. (2018)).

How these differences map into wages identifies the extend of rent-sharing (Guiso et al. (2005); Kline at al. (2019); Lamadon et al. (2020)).

So why a "union premium"?

- Firm selection: Unions are present in highly productive firms where there is potentially a high surplus to be shared with workers.
- Rent extraction: Unions help extract a higher share from a given surplus.

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# ECONOMIC IMPACTS OF NEW UNIONIZATION ON PRIVATE SECTOR EMPLOYERS: 1984–2001*

JOHN DINARDO AND DAVID S. LEE

Economic impacts of unionization on employers are difficult to estimate in the absence of large, representative data on establishments with union status information. Estimates are also confounded by selection bias, because unions could organize at highly profitable enterprises that are more likely to grow and pay higher wages. Using multiple establishment-level data sets that represent establishments that faced organizing drives in the United States during 1984–1999, this paper uses a regression discontinuity design to estimate the impact of unionization on business survival, employment, output, productivity, and wages. Essentially, outcomes for employers where unions *barely* won the election (e.g., by one vote) are compared with those where the unions *barely* lost. The analysis finds small impacts on all outcomes that we examine; estimates for wages are close to zero. The evidence suggests that—at least in recent decades—the legal mandate that requires the employer to bargain with a certified union has had little economic impact on employers, because unions have been somewhat unsuccessful at securing significant wage gains.

Source: DiNardo and Lee (2004)

# DiNardo and Lee (2004): Very nice example of how to frame a paper with zeroes

**Commanding the null hypothesis: Tell the readers which particular prior you want them to have.**

*It is widely understood that unions impose costs on employers: the most important way is by raising members' wages. They can also impose other costs on employers—by limiting discretion in hiring and firing, for example, and altering the structure of pay across skill groups. These constraints can lead employers to reduce employment, output, or most dramatically, to cease operation altogether. Indeed, these effects are often directly acknowledged by employers and employees alike. During union organizing drives, for example, firms routinely threaten to close a plant if the union drive is successful (Bronfenbrenner 1994), and employees seem to take these threats seriously: the risk of plant closure is cited as the leading cause of union withdrawal from organizing attempts.*

# DiNardo and Lee (2004)

**Framing as non-obvious question: introducing ambigious theoretical relationships to motivate empirics.**

*Are the costs of unionization to employers large or small? Today, in the United States, arguments can be made for either case. On the one hand, conventional estimates suggest that there still exists a sizable union wage premium: demographically similar union workers are paid 15 percent or more than their non-union counterparts. To the extent that employers are sensitive to the price of labor, this may lead to large reductions in employment. On the other hand, there is a broad consensus that in the past three decades, union power in the United States has been on the decline. There has been a decrease in union membership and new organizing activity, high levels of managerial opposition, and increased use of permanent replacement workers. During the 1980s, prominent unions were accepting wage cuts, facing the pressures of the opening of international competition.*

# DiNardo and Lee (2004)

**Explaining why it is a difficult problem.**

*At least two important challenges hinder credible measurement of the causal impacts of unionization on employers. One limiting factor is the absence of large, representative data sets that track establishments over time that also provide information on union status. A second important concern is the fact that unionization is nonrandom. Depending on the correlation between factors associated with unionization and those associated with employment, output, and productivity, the observed correlation between union status and employer outcomes may overstate or understate the true effects of unions. Two competing phenomena may induce opposite selectivity biases. On the one hand, unions may tend to organize at highly successful enterprises that are more likely to survive and grow. On the other, a union organizing drive may be more likely to succeed when a firm is poorly managed, or has faced recent difficulties.*

# DiNardo and Lee (2004)

**Say what you do and highlight your strenghts to persuade the reader.**

*In this paper we present quasi-experimental evidence on the causal effect of unionization on employer business failures/dislocations, employment, output, productivity, and wages, using two large databases representative of US establishments at risk of being unionized. Our analysis is based on the fact that most new unionization occurs as a result of a secret ballot election. By law, if a majority of workers vote in favor of the union, the law requires the management to bargain "in good faith" with the recognized union. This process creates a natural set of comparisons between establishments that faced elections where the union barely won (say, by one vote) and those that faced elections where the union barely lost (by one vote). As in other regression-discontinuity designs, the comparison between near winners and near losers potentially eliminates any confounding selection and omitted variable biases, and allows us to devise credible and transparent estimates of the effect of unions on employer outcomes.*

# DiNardo and Lee (2004)

**Bare-miniminal detail about data and data sources.**

*We report several findings from analyzing data that span the 1984–2001 period, and combine information on elections from the National Labor Relations Board (NLRB), on contract expirations from the Federal Mediation and Conciliation Service (FMCS), on subsequent business survival, employment, and output from a commercial database based on telephone listings (InfoUSA), as well as on employment, wages, output, and productivity in the manufacturing sector from the US Census Bureau's Longitudinal Research Database (LRD).*

# DiNardo and Lee (2004)

**To sell a null result, need to show *something* happened.**

*We first document that the outcome of an NLRB election has a substantial, binding impact on the collective bargaining process, even among close elections. Where they barely win the election, unions are able to maintain their legal recognition over long time horizons; where they barely lose, there is little evidence of subsequent attempts to organize the workplace. Furthermore, unions who barely win have as good a chance of securing a collective bargaining agreement with the employer as those who win the elections by wide margins. And, as expected, unions who barely lose an election have little chance of ever signing such an agreement. These facts show that—statistically speaking—employers face a minimal risk of ever entering collective bargaining negotiations after a union loses a closely contested election.*

# DiNardo and Lee (2004)

**Focus on effect sizes. What can you rule out?**

*This legally mandated shift in the bargaining position of the workers, however, does not lead to significant impacts on a number of employer outcomes. First, union effects on business survival are small—on the order of.01 to.02 on a mean survival rate of .40 over an average of eight years. Second, point estimates of the union impacts on employment, output, and productivity, are statistically insignificant; in the manufacturing sector, they range between 3 and 3 percent for production hours, between 4 and 4 percent for output, and between 2 and 0 percent for output per worker, over one- to fifteen-year horizons.*

# DiNardo and Lee (2004)

**Advance your preferred interpretation of results to tell your students. It is okay that readers disagree with it.**

*One interpretation of these results is that the true employment effects are moderately sized, but cannot be detected by our research design, due to sampling variability in our estimates. An alternative interpretation is that the effects are truly small (e.g.,2 or 3 percent). We favor the latter interpretation for the following reason: our estimates of union wage impacts are small—centered around zero—with enough precision to rule out a 2 percent wage increase for up to seven years after the election. This implies—provided that wage and employment outcomes remain on the employer's labor demand schedule—that the impacts on employment are likely to be small, even assuming relatively large labor demand elasticities.*

# DiNardo and Lee (2004)

**Don't oversell, don't undersell.**

*Based on the evidence, we conclude that—at least in recent decades in the United States—the legal mandate that requires the employer to bargain with a certified union has had little economic impact on employers.*

# DiNardo and Lee (2004)

**Reconcile your findings with the previous literature. D&L highlight the importance of research design, while offering other explanations.**

*The small wage effects that we estimate may appear to be at odds with an enormous literature that has documented substantial union wage premiums. The differing results, however, may be explained by some important differences—other than in research design—in the nature of the data used. First, the modern union wage premium literature typically examines individual-level household survey data, rather than establishment-level data as we do here. Freeman and Kleiner (1990) argue that the latter is more appropriate for directly addressing the direct impacts of a workplace becoming unionized. Indeed, other establishment-level analyses find small or statistically undetectable wage effects (Freeman and Kleiner 1990; LaLonde, Marschke, and Troske 1996). Second, the data contain information on recent unionization (within the past twenty years), while most worker-level data sets possess little information on when the union was formed; estimates derived from those data naturally cannot isolate wage impacts that result from unionization that occurred in recent decades. As noted in Freeman and Kleiner, existing wage differences between union and nonunion workers today average the effects of unions of previous periods and the effects of unionization that occurs today.*

# DiNardo and Lee (2004)

**Stand on the shoulders of giants.**

*Our results may also appear to be at odds with the standard "textbook" treatment of the neoclassical theory of union impacts, which emphasizes the notion of a union as an effective "monopoly" on labor services. There is, however, an older tradition in economics that argues—on a purely theoretical level—that most trade unions are unsuccessful monopolies. Indeed, in his essay, "The Impact of the Union," Friedman (1950) argued that the ability of unions to raise wage rates at that time was somewhat exaggerated, because most unions could not overcome market forces that would tend to keep wages aligned with competitive rates. In a published exchange with Paul Samuelson, Friedman explains his reasoning: "I think if [UAW leader Walter] Reuther were to disregard [pressures to moderate wage demands] and if he were to seek—and for the moment let us suppose he is temporarily successful—very radically raised wages, and if that had the effect of grossly reducing employment within the automobile industry you would find opposition building up that would break the union down. Knowing that in advance and being as smart as you and I, he would avoid such action.*

# DiNardo and Lee (2004)

**Don't reinvent the wheel: when writing, follow the usual structure.**

*The paper is organized as follows. Section II provides some background on the union recognition process and the industrial relations climate in the United States in recent decades. Section III describes different notions of the causal impact of unionization, the regression-discontinuity design for estimating direct impacts of unionization, as well as the identification strategy for assessing indirect, "union threat" effects. We describe the various data sets in Section IV, present the results in Section V, and discuss the findings in relation to the existing literature in Section VI. Section VII concludes.*

# Lee and Mas (2012)

## LONG-RUN IMPACTS OF UNIONS ON FIRMS: NEW EVIDENCE FROM FINANCIAL MARKETS, 1961–1999*

DAVID S. LEE AND ALEXANDRE MAS

We estimate the effect of new private-sector unionization on publicly traded firms' equity value in the United States over the 1961–1999 period using a newly assembled sample of National Labor Relations Board (NLRB) representation elections matched to stock market data. Event-study estimates show an average union effect on the equity value of the firm equivalent to $40,500 per unionized worker, an effect that takes 15 to 18 months after unionization to fully materialize, and one that could not be detected by a short-run event study. At the same time, point estimates from a regression discontinuity design—comparing the stock market impact of close union election wins to close losses—are considerably smaller and close to zero. We find a negative relationship between the cumulative abnormal returns and the vote share in support of the union, allowing us to reconcile these seemingly contradictory findings. *JEL* Codes: J01, J08, J5, J51.

Source: Lee and Mas (2012)

# Event-study reveals ↓ 10% in market value, ≈ $40,500 per unionized worker

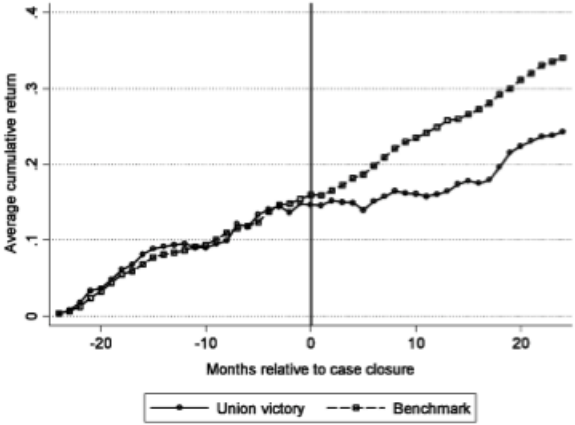


FIGURE II
Average Cumulative Returns of Union Victory Firms and of the Size-Matched Reference Portfolio, by Month Relative to NLRB Case Closure.

# Effects are much smaller and close to zero from RDD

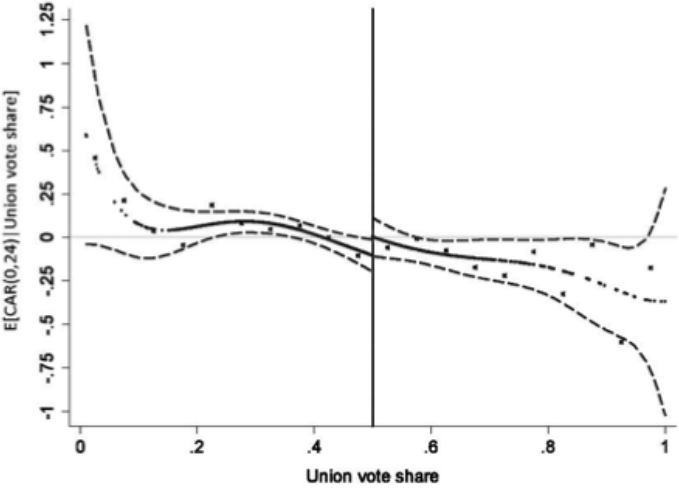


FIGURE VII
Cumulative Abnormal Returns in the Two Years after NLRB Closes Election, by Relation to Vote Share

# Existing evidence on union premium

- Panel data regressions based on workers switching b/w non-union/union job (Card 1996; Lemieux 1998; Angrist and Newey 2012):
  - Akin to lit. of industry wage premium (Krueger and Summers 1988; Gibbons and Katz, 1992)
  - Cannot assess role of firm selection
  - Aggregating moves in either non-union / union sector misses within-union firms pay heterogeneity $\rightarrow$ bias estimates of union wage premiums (Card-Rothstein-Yi, 2023)
- DiNardo and Lee (2004): RD of close elections $\rightarrow$ no wage effects:
  - "Locality" of the RD is crucial here...
  - Frandsen (2021) finds evidence of manipulation in RDs based on union elections
- Lee and Mas (2012): Event-study of winning/losing elections $\rightarrow$ large effects on equity values of firms:
  - No direct evidence on wages + based on firm-level data
  - Can only focus on recent unionization events (Freeman and Kleiner 1990; Sider et al, 2004)

# Beauregard et al. (2024): Why Do Union Jobs Pay More? New Evidence from Matched Employer-Employee Data

- In Canada, collective bargaining is highly centralised:
  - In most cases, all unionised workers in a firm are represented by a single union.
  - But workers in large firms may be represented by different unions (pilots vs. mechanics in airlines) or same union can represent workers from different firms (construction).
- Authors leverage Canada's Business Employee Analytical Microdata (BEAM):
  - Employers must deduct union dues from the wages of each employee in the unit affected by a CBA, regardless of being a union member.
- They estimate employers' average pay policies through AKM approach:
  - Average firm effect across *all* unionised employers, net of worker selection.
  - With firm effects, they decompose union pay premium into firm-selection component and rent-extraction component using Oaxaca-Blinder decomposition.

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# Wage gap between unionized and non-unionized jobs is 15 log points

| | Unionized Jobs | Non-Unionized Jobs (at Non-Union Firm) | Difference |
|---|---|---|---|
| | [1] | [2] | [3] |
| **Panel A: Union Gap** | | | |
| Avg Log Earnings (Residualized) | 0.035 | -0.113 | 0.148 |
| Avg Person Effects | -0.072 | -0.064 | -0.008 |
| Avg Firm Effects | 0.106 | -0.048 | 0.155 |
| Avg Log Value Added per Worker | 1.259 | 0.981 | 0.278 |
| **Panel B: Regression of Firm Effects on Log Value Added per Worker** | | | |
| Constant | -0.172 | -0.259 | 0.087 |
| | (0.00) | (0.00) | (0.00) |
| Slope | 0.221 | 0.215 | 0.006 |
| | (0.00) | (0.00) | (0.00) |
| **Panel C: Oaxaca Decomposition** | | | |
| Rent-Extraction (Levels) | -0.172 | -0.259 | 0.087 |

# But unionized firms' value added is also 28 log points higher

| | Unionized Jobs | Non-Unionized Jobs (at Non-Union Firm) | Difference |
|---|---|---|---|
| | [1] | [2] | [3] |
| **Panel A: Union Gap** | | | |
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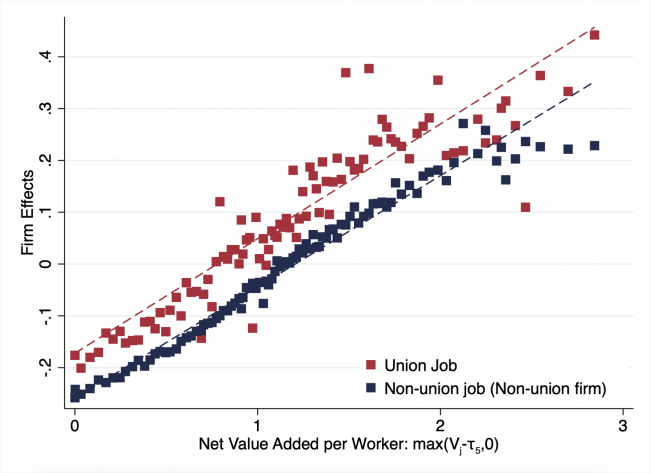
# Unionized firms pay about 9 log points more than non-unionized firms with the same value added per worker




60% of union premium is due to rent-extraction channel and 40% is due to firm selection.

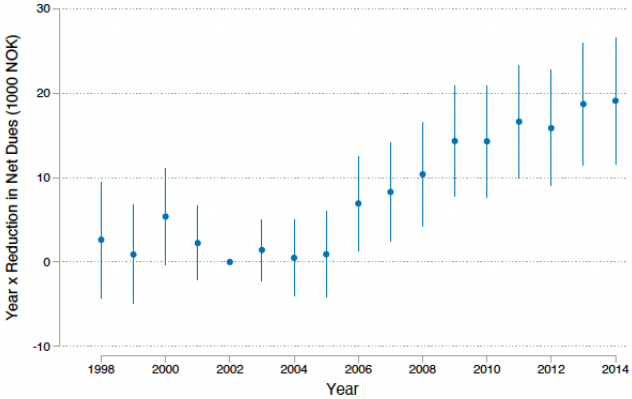
# Roadmap

- What do unions do?
- Why an union pay premium?
- **Beyond pay premium:** Who pays for unions?

# Dodini et al. (2025): Who pays for Unions?

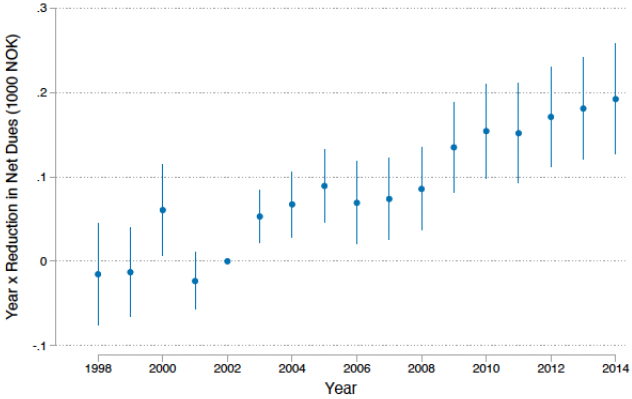
- Context is Norway: A policy change quadrupled the maximum tax deduction for union dues between 2003 and 2010.
- Workers at firms with higher baseline union dues had a stronger incentive to join or remain in union compared to those with lower baseline dues.
- Authors leverage an instrumented dose-response difference-in-differences design, in which union dues subsidy is an instrument for union density.
- The authors focus on manufacturing sector.

# First stage: Firms more exposed to the increased union dues subsidy experience higher union density



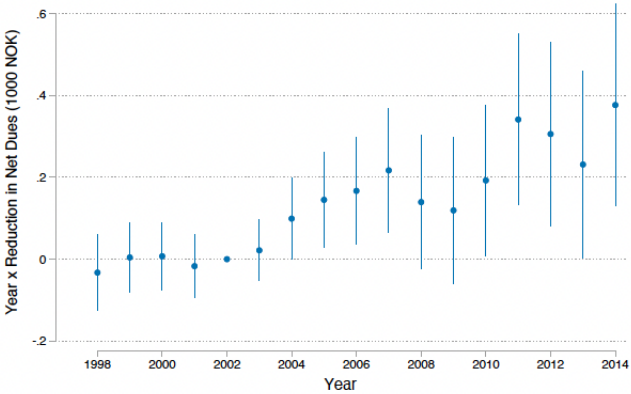

Source: Dodini et al. (2025)

# 1 p.p. increase in union density leads to a 1 log point increase in earnings per worker




Source: Dodini et al. (2025)

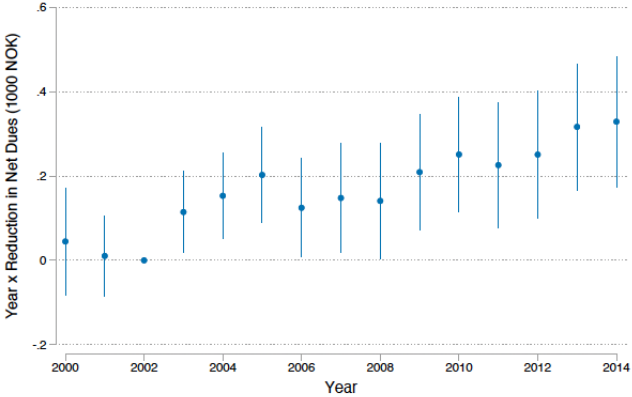
# 1 p.p. increase in union density leads to a 1.1 log point increase in employment




Source: Dodini et al. (2025)

Result is consistent with unions counteracting monopsony power in manufacturing sector.

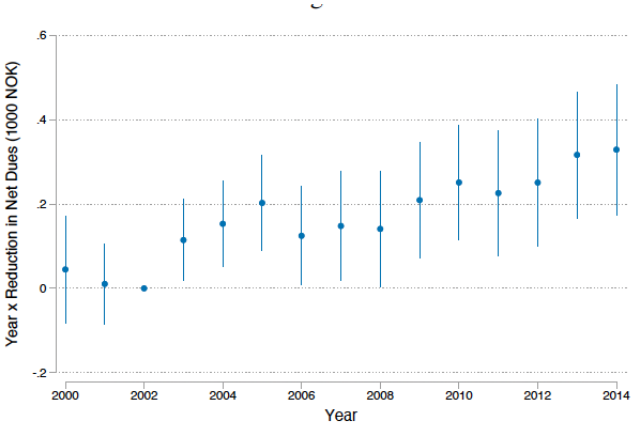
# 1 p.p. increase in union density leads to a 1.1 log point increase in value added per worker




Source: Dodini et al. (2025)

Is the increase in nominal labor productivity driven by increase in quantity produced per worker or increase in price?

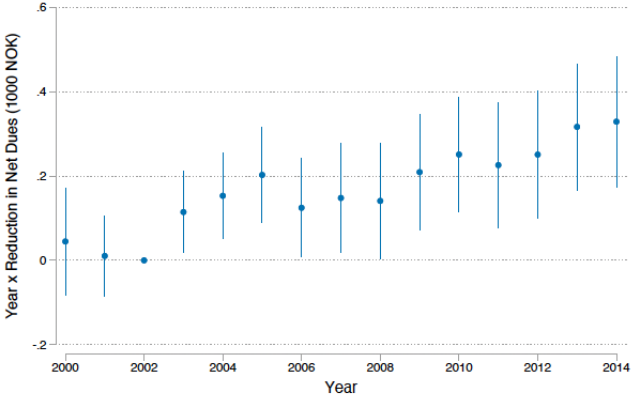
# 1 p.p. increase in union density leads to a 1.1 log point increase in value added per worker




Source: Dodini et al. (2025)

Both! Prices increase, and firms are able to recruit better workers and reduce turnover.

# 1 p.p. increase in union density leads to a 1.1 log point increase in value added per worker

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Three quarters of the increased labor cost from unions is paid for by consumers in the form of higher prices, and the remainder pays for itself in the form of improved productivity.

# Roadmap

- What do unions do?
- Why an union pay premium?
- **Beyond pay premium:** Inequality

# Farber et al. (2021): Unions and inequality over the 20$^{th}$ century: New evidence from survey data

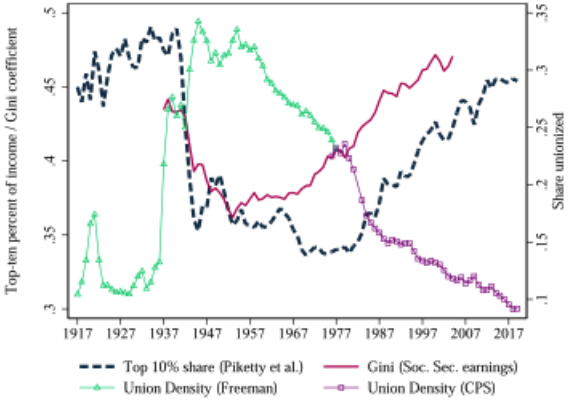


FIGURE I

Union Density and Inequality Measures, 1917–2019

Source: Farber et al. (2021)

# In its mid-century peak, union HHs were much less educated, whereas pre- WWII and today they are more similar to nonunion HHs

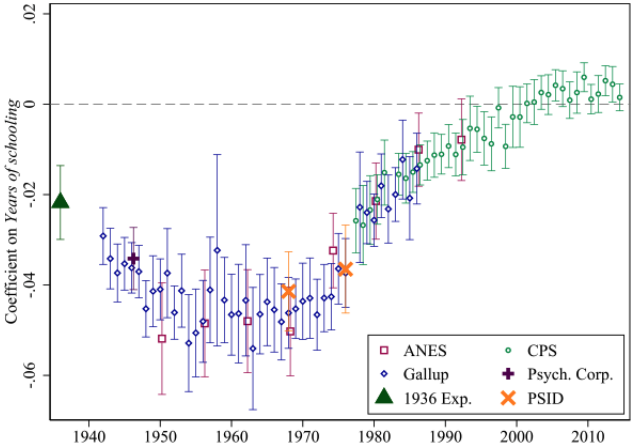


FIGURE III

How Does Years of Schooling Predict Union Household Status?

# In its mid-century peak, union HHs were much more nonwhite, whereas pre- WWII and today they are more similar to nonunion HHs

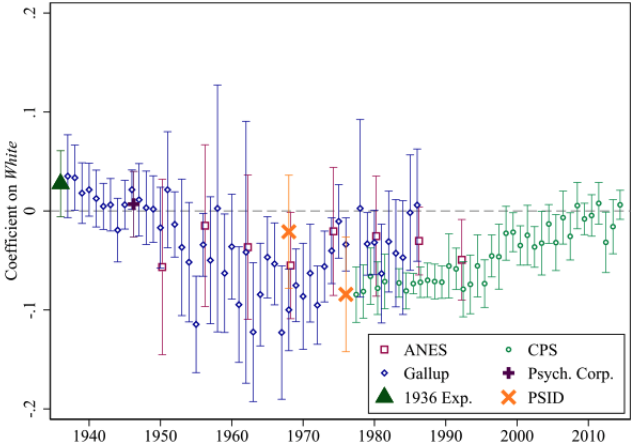


FIGURE IV

How Does Race Predict Union Household Status?

# HH union premium holds relatively steady bw. 10–20 log points

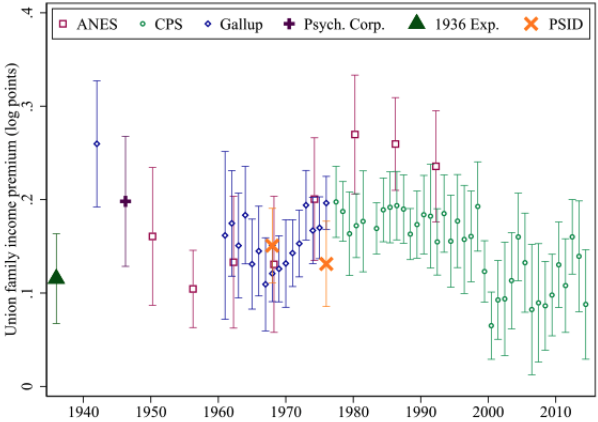


FIGURE V

Estimates of the Union Family Income Premium

Source: Farber et al. (2021)

# If you are particularly interested in unions more generally...

Other firm responses to unions:

- Lagos (2025): Union bargaining power and the amenity-wage tradeoff

What do unions do?

- Corradini et al. (2024): Collective bargaining for women: how unions can create female-friendly jobs?
- Derenoncourt et al. (2025): What do (thousands of) unions do? Union-specific pay premia and inequality
- Dodini et al. (2025): Union membership and native-immigrant labor market gaps

Who are the union leaders?

- Boudreau et al. (2024): Leaders in social Movements: evidence from unions in Myanmar (forthcoming, AER)